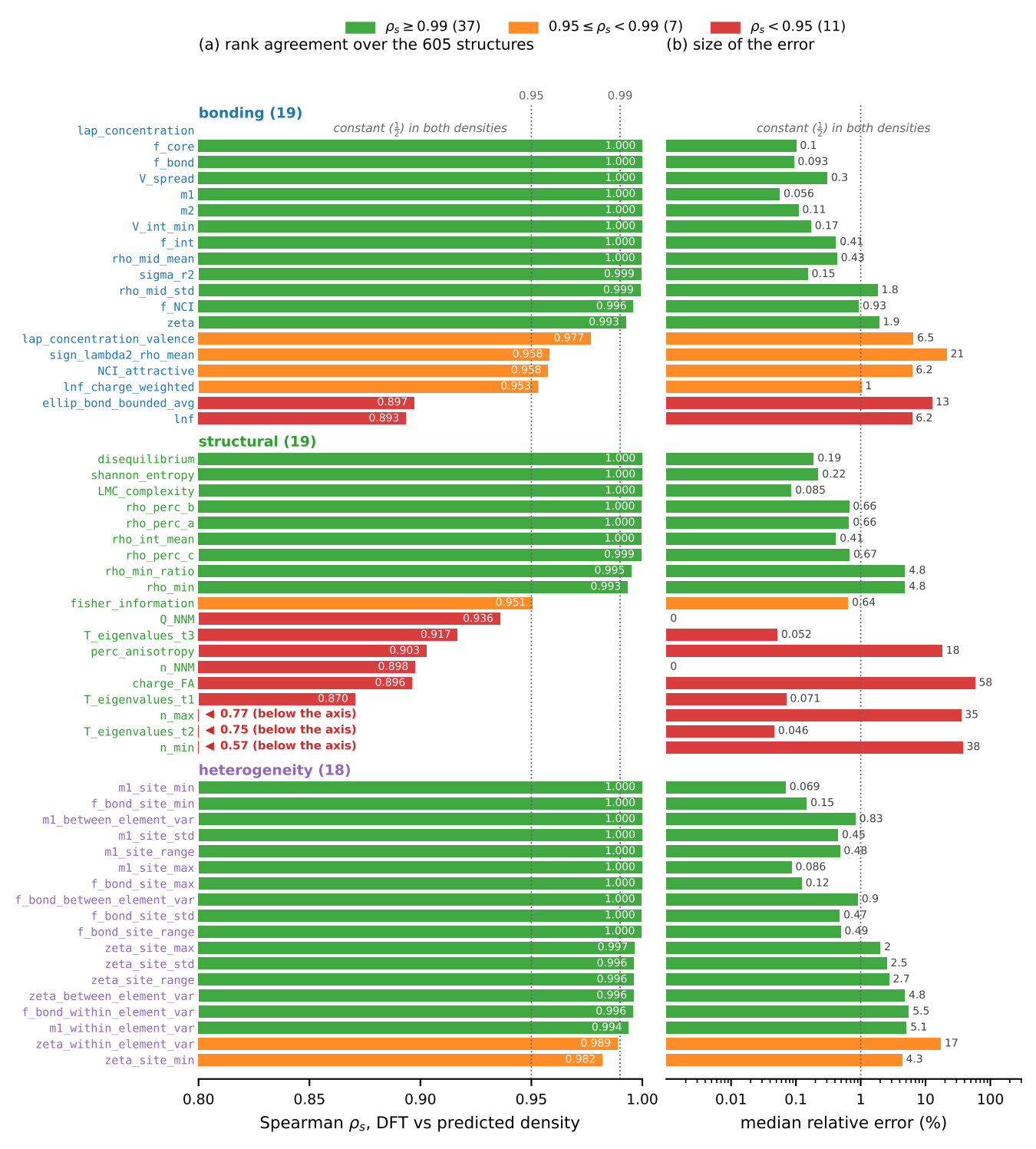
$\rho_s \geq 0.99$ (37)
$0.95 \leq \rho_s < 0.99$ (7)
$\rho_s < 0.95$ (11)
(a) rank agreement over the 605 structures
(b) size of the error
0.95
0.99
bonding (19)
lap_concentration
constant ($\frac{1}{2}$) in both densities
f_core 1.000 0.1
f_bond 1.000 0.093
V_spread 1.000 0.3
m1 1.000 0.056
m2 1.000 0.11
V_int_min 1.000 0.17
f_int 1.000 0.41
rho_mid_mean 1.000 0.43
sigma_r2 0.999 0.15
rho_mid_std 0.999 1.8
f_NCI 0.996 0.93
zeta 0.993 1.9
lap_concentration_valence 0.977 6.5
sign_lambda2_rho_mean 0.958 21
NCI_attractive 0.958 6.2
lnf_charge_weighted 0.953 1
ellip_bond_bounded_avg 0.897 13
lnf 0.893 6.2
structural (19)
disequilibrium 1.000 0.19
shannon_entropy 1.000 0.22
LMC_complexity 1.000 0.085
rho_perc_b 1.000 0.66
rho_perc_a 1.000 0.66
rho_int_mean 1.000 0.41
rho_perc_c 0.999 0.67
rho_min_ratio 0.995 4.8
rho_min 0.993 4.8
fisher_information 0.951 0.64
Q_NNM 0.936 0
T_eigenvalues_t3 0.917 0.052
perc_anisotropy 0.903 18
n_NNM 0.898 0
charge_FA 0.896 58
T_eigenvalues_t1 0.870 0.071
n_max ◀ 0.77 (below the axis) 35
T_eigenvalues_t2 ◀ 0.75 (below the axis) 0.046
n_min ◀ 0.57 (below the axis) 38
heterogeneity (18)
m1_site_min 1.000 0.069
f_bond_site_min 1.000 0.15
m1_between_element_var 1.000 0.83
m1_site_std 1.000 0.45
m1_site_range 1.000 0.48
m1_site_max 1.000 0.086
f_bond_site_max 1.000 0.12
f_bond_between_element_var 1.000 0.9
f_bond_site_std 1.000 0.47
f_bond_site_range 1.000 0.49
zeta_site_max 0.997 2
zeta_site_std 0.996 2.5
zeta_site_range 0.996 2.7
zeta_between_element_var 0.996 4.8
f_bond_within_element_var 0.996 5.5
m1_within_element_var 0.994 5.1
zeta_within_element_var 0.989 17
zeta_site_min 0.982 4.3
0.80
0.85
0.90
0.95
1.00
Spearman ρ_s, DFT vs predicted density
0.01
0.1
1
10
100
median relative error (%)